

ROS 2 + iRobot Create 3 Setup on RPi5

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Resumen

ROS 2 Jazzy was installed on the Raspberry Pi 5 (Ubuntu Server 24.04), and working communication with the iRobot Create 3 mobile base was established, resolving apt package conflicts, a DDS interoperability issue, and an undocumented namespace (`/r1/`).

Installing ROS 2 Jazzy

```
# Locale
sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8

# Universe repository
sudo apt install software-properties-common
sudo add-apt-repository universe

# ROS 2 key and repo
sudo apt install curl -y
sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key
-o /usr/share/keyrings/ros-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-
by=/usr/share/keyrings/ros-archive-keyring.gpg]
http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo
$UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list >
/dev/null

# Install
sudo apt update
sudo apt upgrade
sudo apt install ros-jazzy-ros-base
sudo apt install ros-dev-tools
sudo rosdep init && rosdep update

# Auto-sourcing
echo "source /opt/ros/jazzy/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

Apt Dependency Conflicts

Ubuntu Noble's ARM64 mirror had several `-dev` packages out of sync with security updates already installed (`.1` point versions). It showed up twice, with different packages each time:

Package	Symptom
<code>liblz4-dev</code> , <code>libzstd-dev</code>	required an exact version of <code>liblz4-1</code> / <code>libzstd1</code> that had already been superseded
<code>acl</code> , <code>libacl1-dev</code>	same pattern with <code>libacl1</code>

Fix applied: instead of patching package by package, a `full-upgrade` synced everything at once:

```
sudo apt update
sudo apt full-upgrade -y
```

This allowed `ros-jazzy-ros-base` to install, and later `ros-jazzy-rmw-cyclonedds-cpp` with no further conflicts.

Connecting to the Create 3 and the `/r1/` namespace

`ros2 topic list` initially showed the Create 3's topics with no namespace (`/odom` , `/cmd_vel` , `/battery_state` , etc.), but after reconnecting it turned out the robot actually publishes them under `/r1/` (`/r1/odom` , `/r1/cmd_vel` , ...). This explains why the first attempts to move the robot with `ros2 topic pub /cmd_vel ...` had no effect: the topic didn't match the real one.

Actions discovered with `ros2 action list`:

```
/r1/audio_note_sequence
/r1/dock
/r1/drive_arc
/r1/drive_distance
/r1/led_animation
/r1/navigate_to_position
/r1/rotate_angle
/r1/undock
/r1/wall_follow
```

Reading the Create 3's own message types (`DockStatus` , `HazardDetectionVector` , etc.) requires the `irobot_create_msgs` package, which isn't included in `ros-jazzy-ros-base`:

```

sudo apt install ros-jazzy-irobot-create-msgs
# if no binary exists for Jazzy on this architecture, build from source:
mkdir -p ~/create3_ws/src
cd ~/create3_ws/src
git clone https://github.com/iRobotEducation/irobot_create_msgs.git
cd ~/create3_ws
rosdep install --from-paths src -i -y
colcon build
source install/setup.bash

```

Cyclone DDS instead of Fast DDS

Sending an action (`/undock`) produced:

```
[RTPS_READER_HISTORY Error] Change payload size of '32' bytes is larger than
the history payload size of '19' bytes and cannot be resized.
```

This is an interoperability issue: the Create 3 uses Cyclone DDS internally, while Jazzy defaults to Fast DDS.

Fix:

```

sudo apt install ros-jazzy-rmw-cyclonedds-cpp
export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp
echo "export RMW_IMPLEMENTATION=rmw_cyclonedds_cpp" >> ~/.bashrc
source ~/.bashrc

```

Note: when opening a new SSH session, confirm `~/.bashrc` still contains both the ROS sourcing line and this export (it was lost once after a reconnect).

Control Commands (with the `/r1/` namespace)

Action	Command
Move (one-off teleop)	<pre>ros2 topic pub /r1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.2}, angular: {z: 0.0}}" -r 10</pre>
Keyboard	<pre>ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -r cmd_vel:=/r1/cmd_vel</pre>
Undock	<pre>ros2 action send_goal /r1/undock irobot_create_msgs/action/Undock "{}"</pre>

Action	Command
Return to dock	<code>ros2 action send_goal /r1/dock</code> <code><correct_type> "{}"</code> — exact type still needs confirming (see next section)
List topics	<code>ros2 topic list</code>
List actions	<code>ros2 action list</code>

Important: `--once` on `ros2 topic pub` is not reliable with the Create 3 (it can miss the DDS discovery window) — better to use `-r <Hz>` and Ctrl+C to stop.

Pending / Next Steps

- ☐ Confirm the exact action type for `/r1/dock` with `ros2 action info /r1/dock -t` (the `DockServo` type used initially was wrong)
- ☐ Verify that `/r1/dock` correctly returns the robot to the dock once the type is fixed
- ☐ Decide whether to bake the `/r1/` namespace into AURA_V2's launch files, or rename the robot to drop the namespace from its web interface
- ☐ Integrate the Create 3 as AURA_V2's mobile base (RPLIDAR, STM32, Nav2)